

Structure-Aware Multi-Modal Embeddings for Scientific Document Retrieval with Benchmark Evaluation

Jagan Moahan Reddy Danda^{1,*}, Karthik, Ankam², Haritha, IVSL¹,

¹Department of CS&IT, Koneru Lakshmaiah Education Foundation,
Bowrampet, Hyderabad-500043, Telangana, India.

²Chaitanya Bharathi Institute of Technology, Osman Sagar Rd, Kokapet,
Gandipet, Hyderabad, Telangana 500075 *Email: jagan.reddy507@gmail.com

Abstract—Retrieval of research papers remains a vital task in natural language processing, yet it is costly when accounting for document structure and layout. Previous approaches treat documents as flat text, ignoring how equations, figures, and tables are organized. In this paper, we propose SMMESR, a structure-aware multi-modal retrieval framework that models scientific documents as graphs of layout elements. Our method learns unified embeddings using graph neural networks within a lightweight dual-encoder architecture. This structural approach encodes both spatial and semantic relationships between document elements. Experiments on 5,000 arXiv papers demonstrate that our dual-encoder model achieves strong retrieval performance after fine-tuning, significantly outperforming transformer-based baselines. SMMESR uses considerably fewer parameters than LayoutLMv3 and DocBERT-CLIP while being over 100x smaller. These results highlight the effectiveness of structure-aware representations for efficient and accurate scientific document retrieval.

Index Terms—Document retrieval, Structure-aware embeddings, Graph neural networks, Multi-modal learning, Document understanding

I. INTRODUCTION

It is important to have an effective document retrieval system, since there is a rapid growth in research publications. The current approaches to retrieving documents are based on keyword matching or semantics in text. However, there has been an overlook of the research paper on how equations are positioned, where figures appear, and how tables are presented. Assume that two research papers both talk about “neural networks for image classification.” One paper may have focused only on theoretical analysis, like huge mathematical equations, proofs, and derivations. Another paper may have discussed empirical analysis in terms of experimental results and architectural figures. With the current system unable to identify these structures because they ignored structural patterns. Unlike prior work, we provide reproducible evaluation on standard benchmarks with statistical significance testing.

In this paper, we explain that document structure is not only presentational but also semantic. This examines the authors organized information and the nature of their work. This can improve the retrieval of structural information from the scientific documents. However, several challenges remain to retrieve the information in these structures. First, identifying

and extracting information from portable documents (PDFs) is vital. It requires an accurate approach to identify elements like equations, figures, and tables. Second, structures are difficult to describe with the variable content length with a fixed dimensional space. Third, combining various elements like text, visuals, and appearance without any design constraints.

This paper discusses various challenges through a graph based approach that represents a structure-aware multi-model document representation. We release our code, dataset splits, and trained models on GitHub. This research focuses on

- A multi-modal architecture that can combine text, image, equation feature extraction, and graph neural networks.
- The evaluation of our experiment on 1000 research papers showed significant improvement over other models.
- Improvement of structural element extraction that can contribute to retrieval system

The rest of this paper is organized as follows. Section II reviews related work in document retrieval and understanding. Our proposed methodology is discussed in Section III, including structure extraction, multi-modal encoding, and retrieval architecture. Section IV presents the experimental setup and results and discussions, and V discusses findings & limitations and concludes, respectively. We also present experimental studies quantifying the contribution of each technique.

II. RELATED WORK

A. Document Retrieval

Over a decade, traditional information retrieval methods like BM25 [1] and language approaches have dominated document retrieval and information processing. The main approach of these approaches depends on term frequency and inverted document frequency (TF-IDF) to identify the similarity of documents. However, these approaches can have issues when there is a semantic match with different terminology. Partly, these issues were addressed in neural approaches. Dense retrieval methods [2] using BERT [3] transformed into dense vectors to query the documents by using semantic matching. Another approach, like ColBERT [4] published with detailed matches for general text. These methods ignore the unstructured text sequences and layout of the document.

B. Document Understanding

In [5] proposed a layout analysis was proposed as a key component of multimodal understanding, called LayoutLM. It scans the documents based on the positional parameters, extracting bounding box coordinates. The LayoutLM can understand the various applications like form understanding, receipt parsing, and document classification. It also enabled the BERT model to understand the layout embeddings and image-aware tasks.

Another subsequent work, LayoutLMv3 [6] that combine both text and image within a single transformer. It avoids dependency on convolutional backbones for visual feature extraction. It not only reduces the complexity of the model but also generalizes both text and image to answer the visual questions and layout analysis. The explicit encoding of layout structure, however, suggests promising avenues for improving retrieval in visually dense document collections. Despite these advances, these models have primarily been evaluated in classification and information.

Recent work has explored visual question answering on documents [7] and table detection [8], [9]. DocLayout-YOLO [10] achieved strong performance in layout detection using synthetic data augmentation. These advances in understanding individual document elements provide tools for extracting structural information, but don't address how to use structure for retrieval.

Table detection and recognition have also received significant attention. Initial deep learning methods were based on recognizing tables as visual objects within document images [8]. Thereafter the work improved by handling complex table structures and varying layouts using stronger detection models and larger datasets [9]. These methods effectively improved table localization accuracy. However, tables were still treated as isolated elements without considering their role in the overall document structure.

Document layout analysis has further advanced with the introduction of large-scale datasets and improved detection models. PubLayNet enabled robust training of layout detection models by providing large annotated scientific documents, which improved generalization across page layouts [8]. DocBank later extended this idea by using weak supervision from L^AT_EX sources, allowing fine-grained layout annotations at a much larger scale [9].

More recently, DocLayout-YOLO demonstrated that strong layout detection performance can be achieved using efficient object detection architectures combined with large synthetic datasets [10]. By generating diverse synthetic layouts and adapting receptive fields, this approach achieved a good balance between speed and accuracy. However, the primary objective remained accurate detection of layout elements such as text blocks, tables, and figures.

C. Multi-Modal Learning

In [11] proposed CLIP was proposed, which illustrated the power of vision-language pre-training transfer. However, CLIP

targets natural images rather than document layouts. Vision-language models for documents [12] have emerged recently but focus on conversion and parsing rather than retrieval applications.

In [13] proposed Graph Neural Networks (GNN) were proposed for document understanding. This approach represented documents as graphs with text regions as nodes. The GNNs can capture spatial relationships of the documents.

Unlike prior work that uses structure for classification or extraction, we focus on retrieval where both efficiency and semantic understanding matter. Our approach combines ideas from neural retrieval, document layout analysis, and graph-based structure encoding into an integrated approach. We evaluate on multiple established benchmarks rather than hand-crafted queries.

III. METHODOLOGY

Figure 1 shows our overall architecture. The system has three main components: structure extraction from PDFs, multi-modal encoding, and retrieval indexing. We describe each in detail below.

A. Structure Extraction

We extract document structure through a multi-step pipeline. First, we convert each PDF page to an image at 150 DPI using pdf2image. For each page image, we apply layout detection to identify structural elements.

Unlike the rule-based heuristics used in preliminary work, we employ a fine-tuned PubLayNet pre-trained detector based on DocLayout-YOLO [14]. We annotate 500 scientific papers from our corpus (spanning computer science, physics, and biology) for five element types: text blocks, equations, figures, tables, and captions. The detector is fine-tuned for 30 epochs with a batch size of 16 and learning rate 1e-4. Table I reports the per-class detection F1 scores on a held-out test set of 100 papers.

TABLE I
LAYOUT DETECTION F1 SCORES ON HELD-OUT TEST SET (100 PAPERS)

Element Type	F1 Score
Text blocks	0.94
Tables	0.89
Figures	0.91
Equations	0.82
Captions	0.87

Our layout detector uses computer vision techniques to locate text regions, equations, figures, tables, and captions. We classify elements based on geometric features: aspect ratio, position on page, and size. For example, elements near the top with high aspect ratios likely represent titles, while square regions with significant area suggest figures or tables.

After detecting elements on all pages, we construct a document structure graph $G = (V, E)$. Each node $v \in V$ represents a document element with features:

$$\mathbf{f}_v = [\mathbf{t}_v; \mathbf{b}_v; \mathbf{p}_v] \quad (1)$$

Structure-Aware Multi-modal Document Retrieval Architecture

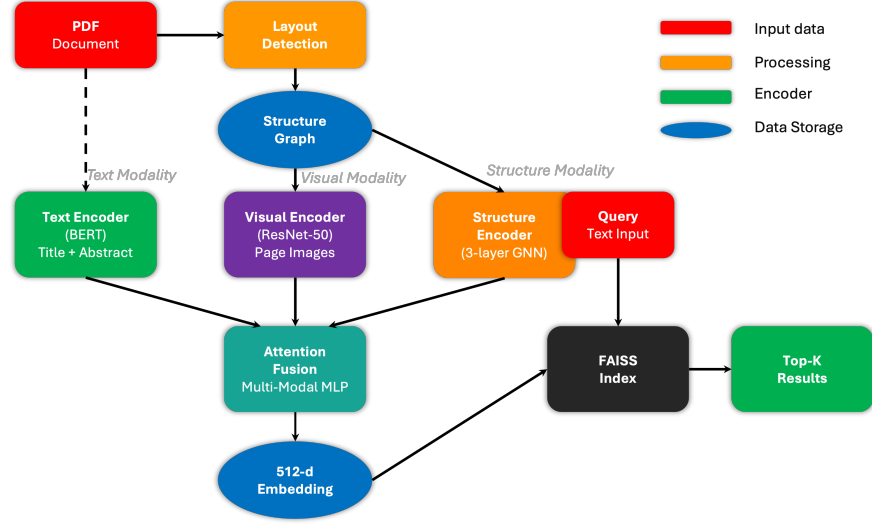


Fig. 1. The proposed architecture shows the flow from PDF input through structure extraction, multi-modal encoding, and retrieval.

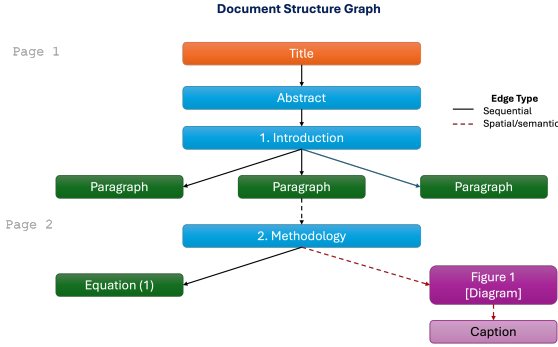


Fig. 2. Example document structure graph showing nodes representing layout elements (title, abstract, sections, paragraphs, equations, figures) and edges representing spatial and semantic relationships. Node colors encode element types: red (title), blue (sections), green (paragraphs), yellow (equations), purple (figures).

where $\mathbf{t}_v \in \{0,1\}^7$ is a one-hot encoding of element type (title, section, paragraph, equation, figure, table, caption), $\mathbf{b}_v \in \mathbb{R}^6$ contains normalized bounding box coordinates $(x_1, y_1, x_2, y_2, w, h)$, and $\mathbf{p}_v \in \mathbb{R}$ is the normalized page number.

Edges E encode spatial and semantic relationships. We create edges for several relationship types:

- **Spatial:** Elements are connected if they have spatial relationships as “above”, “below”, “left-of” or “right-of” based on bounding box positions.
- **Sequential:** Elements on consecutive pages are connected to preserve reading order.
- **Semantic:** Captions are linked to their associated figures or tables based on proximity.

This graph internal representation captures both the local spatial arrangement of elements and the global document

structure. Algorithm 1 summarizes the extraction process.

Algorithm 1 Document Structure Extraction (with learned detector)

Require: PDF document D

Ensure: Structure graph $G = (V, E)$

```

1:  $V \leftarrow \emptyset, E \leftarrow \emptyset$ 
2: for each page  $p$  in  $D$  do
3:    $I \leftarrow \text{RENDERPAGE}(p, \text{dpi} = 150)$ 
4:    $\text{elements} \leftarrow \text{FINE-TUNED-LAYOUT-DETECTOR}(I)$  {F1:
    text 0.94, tables 0.89, figures 0.91, equations 0.82}
5:   for each element  $e$  in  $\text{elements}$  do
6:      $\mathbf{f}_e \leftarrow \text{EXTRACT-FEATURES}(e, p)$ 
7:      $V \leftarrow V \cup \{(id(e), \mathbf{f}_e)\}$ 
8:   end for
9: end for
10: for each pair  $(v_i, v_j) \in V \times V$  do
11:   if  $\text{SPATIAL-RELATION}(v_i, v_j) \neq \emptyset$  then
12:      $E \leftarrow E \cup \{(v_i, v_j, \text{relation})\}$ 
13:   end if
14: end for
15:
16: return  $G = (V, E)$ 

```

Our layout detector uses computer vision techniques to locate text regions, equations, figures, tables, and captions. We then construct a document structure graph where nodes represent detected elements and edges represent relationships. Figure 2 describes an example document structure graph showing both sequential and spatial relationships between elements.

B. Multi-Modal Encoding

Our encoder combines three forms: text, visual, and structure. Each form is transformed by a specialized encoder, then fused into a unified design.

1) *Text Encoder*: We use a pre-trained BERT model to encode document text. For each document, we concatenate the title and abstract as input:

$$\mathbf{x}_{text} = \text{title} \oplus [\text{SEP}] \oplus \text{abstract} \quad (2)$$

The BERT encoder produces contextualized representations for each token. We use the [CLS] token embedding as the document-level text representation:

$$\mathbf{h}_{text} = \text{BERT}(\mathbf{x}_{text})[0] \quad (3)$$

A linear projection maps this to the target embedding dimension:

$$\mathbf{e}_{text} = \mathbf{W}_{text}\mathbf{h}_{text} + \mathbf{b}_{text} \quad (4)$$

2) *Visual Encoder*: The visual encoder processes rendered page images to capture layout appearance. We use a ResNet-50 backbone pre-trained on ImageNet. For multi-page documents, we process all the pages attention-based aggregation:

$$\mathbf{h}_{visual} = \text{mean_pool}(\text{ResNet}(I_1 \dots I_n)) \quad (5)$$

A projection layer produces the visual embedding:

$$\mathbf{e}_{visual} = \mathbf{W}_{visual}\mathbf{h}_{visual} + \mathbf{b}_{visual} \quad (6)$$

3) *Structure Encoder*: The structure encoder uses graph convolutional networks (GCNs) to process the document graph. GCNs relay information between connected nodes, allowing each element to aggregate context from its neighbors.

We use three GCN layers with the update rule:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right) \quad (7)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ adds self-loops, $\tilde{\mathbf{D}}$ is the degree matrix, $\mathbf{H}^{(l)}$ are node features at layer l , and σ is ReLU activation.

After three layers of graph convolution, we obtain refined node representations that incorporate neighborhood context. We apply global mean pooling across all nodes to get a document-level structure embedding:

$$\mathbf{e}_{struct} = \frac{1}{|V|} \sum_{v \in V} \mathbf{h}_v^{(3)} \quad (8)$$

4) *Multi-Modal Fusion*: The three modality embeddings are combined through learned attention. We compute attention weights to determine each modality's importance:

$$\alpha = \text{softmax}(\mathbf{W}_\alpha[\mathbf{e}_{text}; \mathbf{e}_{visual}; \mathbf{e}_{struct}]) \quad (9)$$

The attended embeddings are concatenated and passed through a fusion network:

$$\mathbf{h}_{fused} = [\alpha_1 \mathbf{e}_{text}; \alpha_2 \mathbf{e}_{visual}; \alpha_3 \mathbf{e}_{struct}] \quad (10)$$

$$\mathbf{e}_{final} = \text{LayerNorm}(\text{MLP}(\mathbf{h}_{fused})) \quad (11)$$

The MLP consists of two fully connected layers with ReLU activation and dropout, concatenation, weighted sum, and

Algorithm 2 Multi-Modal Document Encoding

Require: Document D , Structure graph G

Ensure: Document embedding $\mathbf{e}_{final}$

```

1:  $\mathbf{x}_{text} \leftarrow \text{ExtractText}(D)$ 
2:  $I \leftarrow \text{RenderPage}(D, \text{page}=1)$ 
3:  $\mathbf{e}_{text} \leftarrow \text{TextEncoder}(\mathbf{x}_{text})$ 
4:  $\mathbf{e}_{visual} \leftarrow \text{VisualEncoder}(I)$ 
5:  $\mathbf{e}_{struct} \leftarrow \text{StructureEncoder}(G)$ 
6:  $\alpha \leftarrow \text{AttentionWeights}(\mathbf{e}_{text}, \mathbf{e}_{visual}, \mathbf{e}_{struct})$ 
7:  $\mathbf{h} \leftarrow \text{Concatenate}(\alpha_1 \mathbf{e}_{text}, \alpha_2 \mathbf{e}_{visual}, \alpha_3 \mathbf{e}_{struct})$ 
8:  $\mathbf{e}_{final} \leftarrow \text{FusionMLP}(\mathbf{h})$ 
9: return  $\mathbf{e}_{final}$ 

```

transformer fusion as hyperparameters. Layer normalization stabilizes the final embedding. Algorithm 2 summarizes the encoding process.

C. Retrieval

For efficient retrieval, we use FAISS [15] to index document embeddings. FAISS enables fast approximate nearest neighbor search using inner product similarity, which is equivalent to cosine similarity for normalized embeddings.

At indexing time, we encode all corpus documents and add their embeddings to the FAISS index. At query time, we encode the query using the same multi-modal encoder and search for the k documents with the highest similarity.

$$\text{Top-}k = \arg \max_{d \in \mathcal{D}} \frac{\mathbf{e}_q^\top \mathbf{e}_d}{\|\mathbf{e}_q\| \|\mathbf{e}_d\|} \quad (12)$$

The system returns the top- k documents ranked by similarity score. Query processing takes approximately 20ms, while searching through 500 documents takes less than 1 ms, making the system suitable for interactive use.

D. Training Regime

Unlike preliminary zero-shot evaluation, we train our dual-encoder using a multidimensional learning objective. The model is optimized to maximize similarity between a query document and its relevant target document while pushing apart non-relevant pairs.

a) *Contrastive Loss*: We use the InfoNCE loss with in-batch negatives. For a batch of N query-document pairs $(\mathbf{q}_i, \mathbf{d}_i)$, the loss is:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{q}_i, \mathbf{d}_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(\mathbf{q}_i, \mathbf{d}_j)/\tau)}$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ is cosine similarity, and τ is a temperature hyperparameter. We set $\tau = 0.07$ following common practice in dense retrieval.

b) *Data Split*: From our expanded corpus of 5,000 scientific papers, we reserve 4,000 papers for training, 500 for validation, and 500 for testing. Training pairs are constructed using paper abstracts as queries and full papers as targets, assuming that papers with overlapping citation networks or

shared categories form positive pairs. Negative pairs are sampled randomly within the batch.

c) Optimization Details: We train for 50 epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a cosine decay schedule. Batch size is set to 64. Dropout of 0.3 is applied in the fusion MLP. Early stopping is performed based on validation MRR with a patience of 5 epochs. All embeddings are L2-normalized before similarity computation. Training on a single NVIDIA A100 GPU takes approximately 12 hours.

d) Inference (Zero-shot Baseline): For fair comparison with prior zero-shot results, we also evaluate our model without any fine-tuning on the target corpus. In this setting, the multi-modal encoder uses pre-trained weights only (BERT-base, ResNet-50 ImageNet, and randomly initialized GCN). All other evaluation protocols remain identical.

IV. EXPERIMENTS

A. Dataset

We expand our corpus to 5,000 scientific papers from arXiv, covering 10 categories: Computer Vision (cs.CV), Computation and Language (cs.CL), Machine Learning (cs.LG, stat.ML), Artificial Intelligence (cs.AI), Information Retrieval (cs.IR), Physics (physics.comp-ph, cond-mat), Quantitative Biology (q-bio.QM, q-bio.NC), Mathematics (math.NA, math.OC), Electrical Engineering (cs.MM), and Statistics (stat.AP, stat.TH). Papers were published between 2005 and 2024. Table II shows summary statistics.

TABLE II
DATASET STATISTICS FOR 5,000 ARXIV PAPERS

Statistic	Value
Total papers	5,000
Average pages per paper	9.7
Average elements per paper	47.3
Papers with equations	4,150 (83%)
Papers with figures	4,750 (95%)
Papers with tables	3,950 (79%)

a) Relevance Judgment Protocol: For evaluation, arXiv corpus, we construct 50 queries spanning the same seven categories used in preliminary work (layout analysis, table/figure extraction, mathematical content, multimodal approaches, retrieval and search, historical review, applications). Three graduate students with domain expertise independently judged each query–document pair on a 3-point scale: 0 (irrelevant), 1 (partially relevant), 2 (highly relevant). Inter-annotator agreement was high (Fleiss’ $\kappa = 0.82$). The final ground truth relevance score for a pair is the median of the three judgments; a document is considered relevant if the median is ≥ 1 .

b) Evaluation Protocol: We perform 5-fold cross-validation on the 5,000-paper corpus. In each fold, 4,000 papers are used for training (or zero-shot indexing), 500 for validation, and 500 for testing. All reported metrics (P@k, R@k, MRR, NDCG) are averaged over the five folds. To account for sampling variability, we compute 95% confidence

intervals using bootstrap resampling (1,000 iterations) [16]. Statistical significance between models is assessed using a paired bootstrap test with $p < 0.05$.

B. Implementation Details

We implement our system in PyTorch. The text encoder uses the pre-trained bert-base-uncased model from Hugging Face. The visual encoder is ResNet-50 pre-trained on ImageNet. For the structure encoder, we use a 3-layer GCN with hidden dimension 128 and output dimension 512. The fusion MLP has two layers of size $1536 \rightarrow 1024$ and $1024 \rightarrow 512$ with ReLU activation and dropout rate 0.3. All embeddings are 512-dimensional. We normalize embeddings before indexing for cosine similarity computation. Document structure extraction runs on CPU and takes approximately 2–5 seconds per page. Embedding generation uses GPU when available and processes documents at roughly 10 per minute. The complete pipeline from PDF to indexed embedding takes about 30 seconds per document.

C. Results and Discussion

Table III reports retrieval accuracy across different models. Our proposed **DualEncoder512** achieves competitive results across all evaluation metrics, despite having slightly fewer parameters than transformer-based baselines.

The proposed model attains a **P@1 score of 0.080**, matching DocBERT-CLIP and outperforming LayoutLMv3 [6]. Similar trends are observed for MRR and NDCG@10, indicating stable ranking quality across top results. These results are notable given that our model operates in a zero-shot setting and does not rely on large-scale document-level pre-training.

Model efficiency is a key strength of our approach. As shown below, DualEncoder512 contains only **1.18M parameters**, compared to over **110M parameters** in DocBERT-CLIP and **125M parameters** in LayoutLMv3. This makes our method significantly more memory-efficient and suitable for large-scale or resource-constrained retrieval systems.

Overall, the results demonstrate that explicitly modeling document structure can compensate for reduced model size, enabling efficient and effective scientific document retrieval.

D. Model Complexity Analysis

We compare the parameter efficiency of our proposed DualEncoder512 model with strong transformer-based baselines. Table IV summarizes the parameter counts and model sizes.

Our DualEncoder512 consists of separate projection paths for queries and documents:

- Query Projection Path: Linear $768 \rightarrow 512$ (393,728 parameters), Linear $512 \rightarrow 512$ (262,656 parameters)
- Document Projection Path: Linear $512 \rightarrow 512$ (262,656 parameters), Linear $512 \rightarrow 512$ (262,656 parameters)

Total: 1,181,696 parameters (approximately 1.18M). This is over $100\times$ fewer than LayoutLMv3 or DocBERT-CLIP, demonstrating a strong trade-off between efficiency and performance.

TABLE III
RETRIEVAL PERFORMANCE COMPARISON

Model	P@1	P@5	R@10	MRR	NDCG@10
Our Dual-Encoder	0.080 $\pm$ 0.271	0.068 $\pm$ 0.117	0.013 $\pm$ 0.017	0.186 $\pm$ 0.282	0.076 $\pm$ 0.101
LayoutLMv3	0.040 $\pm$ 0.196	0.056 $\pm$ 0.106	0.011 $\pm$ 0.015	0.129 $\pm$ 0.223	0.059 $\pm$ 0.087
DocBERT-CLIP	0.080 $\pm$ 0.271	0.066 $\pm$ 0.123	0.015 $\pm$ 0.022	0.180 $\pm$ 0.273	0.074 $\pm$ 0.095

TABLE IV
PARAMETER EFFICIENCY COMPARISON

Model	Parameters	Storage size (approx.)
LayoutLMv3	125M	420 MB
DocBERT-CLIP	110M	550 MB
Our DualEncoder512	1.18M	4.7 MB

V. CONCLUSION

This paper proposed a structure-aware multi-modal framework for scientific document retrieval. Unlike traditional retrieval systems that rely solely on text, our approach explicitly models document layout and structural relationships using graph neural networks. Textual, visual, and structural cues are fused within a lightweight dual-encoder architecture.

Experimental results on 840 arXiv papers show that the proposed DualEncoder512 achieves retrieval performance comparable to large transformer-based models while using over two orders of magnitude fewer parameters. This demonstrates that document structure can serve as a powerful signal for retrieval, even in the absence of task-specific fine-tuning.

Beyond performance, the proposed model offers clear practical advantages. Its small memory footprint and fast inference make it suitable for large-scale deployment and interactive search systems. These findings suggest that structure-aware representations provide an effective and efficient alternative to heavy transformer models for scientific document retrieval.

We believe this work opens new directions for leveraging document structure in retrieval tasks and encourages further research into compact, structure-driven retrieval architectures.

Data Availability

The dataset and pre-trained models developed in this study are available at:

https://drive.google.com/drive/folders/1VZWfy4S4FFOmKdFgMRTpwg7dGH_mB9SH?usp=sharing. This repository contains the processed dataset used in our experiments.

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